

Lenart–Sottile’s chain formula and Samuel’s chain formula are the same functional in two coordinate systems

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Abstract

Question Q345 asked whether the chain formula of Lenart–Sottile for skew Schubert polynomials (`math/0202090`, Theorem 2) is a specialisation of the tensor-algebra chain formula attributed to Samuel (`MO 313951`), proved in `2026-10-04-c4-samuel-chain-formula.tex`. The answer is yes in the direction that was guessed, by an explicit universal integer linear map that does not depend on the permutations involved (Theorem 5.1, proved here from scratch). The guess moreover understates the situation: that map is invertible, so the two formulas are *equivalent* rather than one being a degeneration of the other — though invertibility, unlike the forward direction, leans on one classical fact I cite without re-deriving (Theorem 5.3).

Concretely, both are the single linear functional $\Lambda_{w/u}: f \mapsto \langle \mathfrak{S}_u f, \mathfrak{S}_w \rangle$ on $H^{2m}(\mathbb{F}\ell_n)$, evaluated on two different families of classes: Samuel’s tensor records $\Lambda_{w/u}$ on the *Monk-generated* monomials $Y^\beta = \prod_k (x_1 + \cdots + x_k)^{\beta_k}$, while Lenart–Sottile’s chain counts record it on the *Pieri-generated* basis $h_\alpha = \prod_k h_{\alpha_k}(x_1, \dots, x_k)$. The underlying *sets* of combinatorial objects are literally identical: a labelled chain in the Lenart–Sottile réseau is a Monk-labelled chain, because the second label coordinate is a function of the cover and not an independent choice (Theorem 2.1).

The transition map in the direction asked about is elementary and explicit, and needs no structure theory at all (Theorem 5.1); the converse direction needs that $\{h_\alpha\}_{\alpha \leq \delta}$ is a $\mathbb{Z}$ -basis (Theorem 5.5), which is where the citation enters. All statements are verified exhaustively for $n \leq 4$ and, for the principal identity, for $n = 5$ (53 081 instances, 0 disagreements), against a Schubert instrument built from divided differences that knows nothing about chains.

Three secondary findings. (i) Samuel’s tensor is *symmetric*, $T_{w/u} \in \text{Sym}^m(V)$ (Theorem 3.1) — not visible from its chain definition. (ii) No scalar factor can relate the two formulas, for a structural reason: on a single block the Lenart–Sottile count is multiplicity-free ($\in \{0, 1\}$ in all 6575 tests at $n = 5$) while the Samuel count is not (it takes the values 0, 1, 2, 3) (Proposition 6.1). (iii) The pre-session conjecture that I_α is the content-sum of $N_{w/u}$ is *refuted* (481 of 603 instances at $n = 4$).

1 What the two sides are

1.1 Conventions

Let S_n act on positions, with right multiplication: $u \triangleleft w$ is a Bruhat cover iff $w = u t_{ij}$ for some $i < j$ with $\ell(w) = \ell(u) + 1$, where t_{ij} exchanges positions i and j . Write $s_p = t_{p,p+1}$, $w_0 = n n-1 \cdots 1$, and $\delta = (n-1, n-2, \dots, 1)$. Schubert polynomials $\mathfrak{S}_w \in \mathbb{Z}[x_1, x_2, \dots]$ are normalised by $\mathfrak{S}_{w_0} = x_1^{n-1} \cdots x_{n-1}$ and $\mathfrak{S}_{ws_j} = \partial_j \mathfrak{S}_w$ when $\ell(ws_j) = \ell(w) - 1$; structure constants are $\mathfrak{S}_u \mathfrak{S}_v = \sum_w c_{u,v}^w \mathfrak{S}_w$. We abbreviate

$$Y_p := \mathfrak{S}_{s_p} = x_1 + \cdots + x_p \quad (Y_0 := 0),$$

so that $x_p = Y_p - Y_{p-1}$, and for a composition β with $n - 1$ parts, $Y^\beta := Y_1^{\beta_1} \cdots Y_{n-1}^{\beta_{n-1}}$. For a word $\mathbf{p} = (p_1, \dots, p_m)$ we write $Y_{\mathbf{p}} := Y_{p_1} \cdots Y_{p_m}$, and $\mathbf{p}_\beta := 1^{\beta_1} 2^{\beta_2} \cdots$ for the unique weakly increasing word of content β .

Throughout, $u \leq w$ in S_n and $m = \ell(w) - \ell(u)$.

1.2 Side A: Samuel's chain formula

This is Theorem 1.1 below, proved in 2026-10-04-c4-samuel-chain-formula.tex. Let $W = \bigoplus_{a < b} \mathbb{Q}\tau_{ab}$, let $R = \text{span}\{\tau_{ab} + \tau_{bc} - \tau_{ac} : a < b < c\}$, and $V = W/R$. By Lemma 2.3 of that paper, V is free on $e_i := t_{i,i+1}$ and $t_{ab} = e_a + \cdots + e_{b-1}$; writing α_i for e_i identifies V with the span of the simple roots of type A , with $t_{ab} \mapsto \alpha_a + \cdots + \alpha_{b-1}$. Set

$$T_{w/u} = \sum_{\substack{u t^{(1)} \cdots t^{(m)} = w \\ \text{lengths increase by 1}}} t^{(1)} \otimes \cdots \otimes t^{(m)} \in V^{\otimes m}.$$

Theorem 1.1 (Samuel; proved 2026-10-04). $T_{w/u} = \sum_{\ell(v)=m} c_{u,v}^w T_v$ in $V^{\otimes m}$.

Expanding each t_{ab} in the root basis gives the coordinates of $T_{w/u}$:

$$T_{w/u} = \sum_{\mathbf{p}} N_{w/u}(\mathbf{p}) \alpha_{p_1} \otimes \cdots \otimes \alpha_{p_m}, \quad N_{w/u}(\mathbf{p}) = \# \left\{ \begin{array}{l} u = u_0 \leq u_1 \leq \cdots \leq u_m = w, \quad u_{s-1}^{-1} u_s = t_{i_s j_s}, \\ \text{with } i_s \leq p_s < j_s \text{ for every } s \end{array} \right\}. \quad (1)$$

We call these *Monk-labelled chains*, because $i \leq p < j$ is exactly the label condition in Monk's rule $\mathfrak{S}_{s_p} \mathfrak{S}_u = \sum_{i \leq p < j, \ell(ut_{ij}) = \ell(u)+1} \mathfrak{S}_{ut_{ij}}$. The load-bearing identity behind Theorem 1.1 — equation (**) of that paper, proved there by induction from Monk's rule — is

$$\mathfrak{S}_u \cdot Y_{\mathbf{p}} = \sum_w N_{w/u}(\mathbf{p}) \mathfrak{S}_w. \quad (2)$$

1.3 Side B: Lenart–Sottile's chain formula

All statements in this subsection are quoted from the L^AT_EX source of math/0202090 (*Skew Schubert polynomials*, Lenart–Sottile), fetched and compiled locally so that theorem numbers are resolved rather than counted; the source entry is graded **deep-read**.

The *labelled Bruhat order* is the directed multigraph with an edge

$$u \xrightarrow{(k,b)} w \quad \text{for each cover } u \leq w \text{ with } u^{-1}w = (i, j), \quad i \leq k < j, \quad b = u(i) = w(j),$$

so there are $j - i$ edges per cover. A chain is *increasing* if its sequence of label *pairs* is strictly increasing in lexicographic order. The *type* of a chain is the composition α counting occurrences of each value as a *first* coordinate, and $I_\alpha(u, w)$ is the number of increasing chains from u to w of type α . Finally, for $\alpha \leq \delta$,

$$h_\alpha := h_{\alpha_1}(x_1) h_{\alpha_2}(x_1, x_2) \cdots h_{\alpha_{n-1}}(x_1, \dots, x_{n-1}).$$

Theorem 1.2 (Lenart–Sottile, Theorem 2). $\mathfrak{S}_{w/u}(x) = \sum_\gamma x^\delta / x^\gamma$, over all increasing chains γ from u to w ; equivalently $\mathfrak{S}_{w/u} = \sum_\alpha I_\alpha(u, w) x^{\delta-\alpha}$.

Their engine is Sottile’s Pieri formula [2, eq. (7)], quoted in `math/0202090` as

$$\sigma_u \cdot h_a(x_1, \dots, x_k) = \sum_{\gamma} \sigma_{\text{end}(\gamma)}, \quad (3)$$

the sum over increasing chains from u of length a all of whose labels have first coordinate k .

Remark 1.3 (honest source grading). I have *not* read [2]. Equation (3) is a verbatim quotation from a paper I did read at source, and Lemma 4.2 below, which is the only place it is used, is independently verified against the divided-difference Schubert instrument in §7 (1009/1009 instances at $n = 4$). No step of this paper rests on an unread proof without such a check.

2 The two réseaux are one réseau

The first observation is that the two papers are not merely analogous: they enumerate the same objects.

Theorem 2.1. *Forgetting second coordinates is a bijection*

$$\{\text{labelled chains } u \rightarrow w \text{ in the Lenart–Sottile réseau}\} \xrightarrow{\sim} \{\text{Monk-labelled chains } u \rightarrow w\},$$

compatible with the statistics: a labelled chain with first-coordinate word $\mathbf{p}$ maps to a Monk-labelled chain with label word $\mathbf{p}$. Consequently

$$N_{w/u}(\mathbf{p}) = \#\{\text{labelled chains } u \rightarrow w \text{ with first-coordinate word } \mathbf{p}\}.$$

Proof. A labelled chain is a sequence of edges $u_{s-1} \xrightarrow{(k_s, b_s)} u_s$. By definition of the réseau, the data of an edge out of a given u_{s-1} consists of a cover $u_{s-1} \triangleleft u_s$, say with $u_{s-1}^{-1}u_s = t_{i_s j_s}$, together with a choice of k_s subject to $i_s \leq k_s < j_s$; the second coordinate is then *determined*, $b_s = u_{s-1}(i_s)$. So a labelled chain is precisely a saturated Bruhat chain $u = u_0 \triangleleft \dots \triangleleft u_m = w$ together with a word $\mathbf{p} = (k_1, \dots, k_m)$ satisfying $i_s \leq k_s < j_s$ for each s . That is verbatim the index set of (1). $\square$

So the difference between the two formulas is not in the objects. It is in (a) which subset is kept, and (b) which statistic is recorded.

Theorem 2.2. *Let α be a composition with $|\alpha| = m$. Then*

$$I_{\alpha}(u, w) = \# \left\{ \begin{array}{l} \text{labelled chains } u \rightarrow w \text{ with first-coordinate word } \mathbf{p}_{\alpha}, \\ \text{whose second coordinates are strictly increasing within} \\ \text{each maximal constant block of } \mathbf{p}_{\alpha} \end{array} \right\}.$$

In particular

$$I_{\alpha}(u, w) \leq N_{w/u}(\mathbf{p}_{\alpha}),$$

with equality whenever $\max_k \alpha_k \leq 1$.

Proof. Lexicographic increase of the pairs (k_s, b_s) says exactly: $k_1 \leq \dots \leq k_m$, and whenever $k_s = k_{s+1}$ also $b_s < b_{s+1}$. A weakly increasing word of content α is unique, namely $\mathbf{p}_{\alpha}$; and “ b strictly increasing whenever k is constant” is the stated block condition, since within a maximal constant block transitivity makes the consecutive condition equivalent to strict increase across the block. The inequality follows because the set described is a subset of the set counted by $N_{w/u}(\mathbf{p}_{\alpha})$ in Theorem 2.1. If every $\alpha_k \leq 1$, every block has size 1 and the condition is empty. $\square$

Remark 2.3 (why the obvious guess fails). The natural first guess — that $I_\alpha(u, w)$ is the sum of $N_{w/u}(\mathbf{p})$ over all words $\mathbf{p}$ of content α — is false in both directions at once. By Theorem 2.2 only the *sorted* word contributes, so the sum over-counts by the symmetry of N (Theorem 3.1 below makes this a factor of exactly $\binom{m}{\alpha}$); and even the single sorted term over-counts, because the second-coordinate condition is discarded. Both effects are visible at $n = 4$: see §7.

Remark 2.4 (the strictness in the lexicographic order is not vacuous, but almost is). Replacing strict by weak lexicographic increase changes nothing: two *consecutive* labels of a chain can never be equal. Indeed if $u_{s-1} \xrightarrow{(k,b)} u_s$ with $u_{s-1}^{-1}u_s = t_{ij}$ and $b = u_{s-1}(i)$, then $u_s(j) = b$ with $j > k$, so in u_s the value b sits strictly to the right of position k ; an edge out of u_s with the same label (k, b) would need $b = u_s(i')$ with $i' \leq k$, which is impossible. A repeated label elsewhere in the chain is possible — 598 of the 4436 labelled chains at $n = 4$ contain one — because a cover moves two values and can carry b back to the left; but non-consecutive repeats are already excluded by monotonicity. This was found by a planted control that failed to fire; see §7.1.

3 Samuel’s tensor is symmetric

The following is invisible from the definition of $T_{w/u}$ as a sum over chains, and is what makes “content” the right index on the Samuel side.

Theorem 3.1. $N_{w/u}(\mathbf{p})$ depends only on the content of $\mathbf{p}$. Equivalently, $T_{w/u} \in \text{Sym}^m(V) \subseteq V^{\otimes m}$.

Proof. By (2), $\sum_w N_{w/u}(\mathbf{p}) \mathfrak{S}_w = \mathfrak{S}_u Y_{p_1} \cdots Y_{p_m}$. The polynomial ring is commutative, so the right-hand side depends on $\mathbf{p}$ only through its multiset of entries. Schubert polynomials are linearly independent, so the coefficients $N_{w/u}(\mathbf{p})$ do too. $\square$

Corollary 3.2.
$$\sum_{\text{content}(\mathbf{p})=\alpha} N_{w/u}(\mathbf{p}) = \binom{m}{\alpha} N_{w/u}(\mathbf{p}_\alpha).$$

4 Both formulas are one linear functional

Definition 4.1. For $u \leq w$ with $m = \ell(w) - \ell(u)$ define $\Lambda_{w/u}: H^{2m}(\mathbb{F}\ell_n) \rightarrow \mathbb{Z}$ by letting $\Lambda_{w/u}(f)$ be the coefficient of σ_w in $\sigma_u \cdot f$.

This is $\mathbb{Z}$ -linear, and by the duality of the intersection pairing it is the functional that carries all the structure constants out of (u, w) : if $f = \sum_v a_v \sigma_v$ then $\Lambda_{w/u}(f) = \sum_v a_v c_{u,v}^w$.

Lemma 4.2. $\mathfrak{S}_u \cdot h_\alpha = \sum_w I_\alpha(u, w) \mathfrak{S}_w$, i.e. $I_\alpha(u, w) = \Lambda_{w/u}(h_\alpha)$.

Proof. Apply (3) successively with $k = 1, 2, \dots, n-1$ to the factors of $h_\alpha = \prod_k h_{\alpha_k}(x_1, \dots, x_k)$. This expresses $\sigma_u h_\alpha$ as a sum of σ_{end} over concatenations $\gamma^{(1)}\gamma^{(2)} \cdots \gamma^{(n-1)}$, where $\gamma^{(k)}$ is an increasing chain of length α_k with all first coordinates equal to k . Such a concatenation is increasing as a single chain: within a block it is increasing by hypothesis, and across the junction from block k to block $k' > k$ the first coordinate strictly increases, so the pairs increase lexicographically regardless of the second coordinates. Conversely, an increasing chain of type α has weakly increasing first coordinates (Theorem 2.2), hence splits uniquely into such blocks. So the concatenations are exactly the increasing chains of type α , and there are $I_\alpha(u, w)$ of them ending at w . $\square$

Theorem 4.3. For all $u \leq w$ in S_n with $m = \ell(w) - \ell(u)$,

$$N_{w/u}(\mathbf{p}) = \Lambda_{w/u}(Y_{\mathbf{p}}), \quad I_{\alpha}(u, w) = \Lambda_{w/u}(h_{\alpha}).$$

Thus Samuel's tensor and Lenart–Sottile's chain counts are the same functional $\Lambda_{w/u}$, read on two different families of classes: $\{Y_{\mathbf{p}}\}$, which spans $H^{2m}(\mathbb{F}\ell_n)$, and $\{h_{\alpha}\}_{|\alpha|=m, \alpha \leq \delta}$, which is a basis of it.

Proof. The two displayed identities are (2) and Lemma 4.2. The Y_p generate $H^*(\mathbb{F}\ell_n)$ as a ring, since $x_p = Y_p - Y_{p-1}$; hence the degree- m monomials $Y_{\mathbf{p}}$ span H^{2m} . That $\{h_{\alpha}\}_{\alpha \leq \delta}$ is a basis is Theorem 5.3. $\square$

5 The transition, in both directions

5.1 Lenart–Sottile from Samuel: elementary and explicit

This is the direction Q345 asked about, and it requires nothing beyond substitution.

Theorem 5.1. Define integers $\chi_{\alpha}(\beta)$ by expanding, in $\mathbb{Z}[x_1, \dots, x_{n-1}]$,

$$h_{\alpha} = \sum_{\beta} \chi_{\alpha}(\beta) Y^{\beta}, \quad \text{obtained by substituting } x_p = Y_p - Y_{p-1}.$$

Then for every $u \leq w$ with $\ell(w) - \ell(u) = |\alpha|$,

$$I_{\alpha}(u, w) = \sum_{\beta} \chi_{\alpha}(\beta) N_{w/u}(\mathbf{p}_{\beta}).$$

The coefficients $\chi_{\alpha}(\beta)$ depend on neither u nor w .

Proof. $Y_1, \dots, Y_{n-1}$ are algebraically independent polynomials with $x_p = Y_p - Y_{p-1}$, so every element of $\mathbb{Z}[x_1, \dots, x_{n-1}]$ — in particular each h_{α} , which is a polynomial with integer coefficients — is uniquely an integral polynomial in the Y_p , homogeneous of the same degree. This is an identity in the polynomial ring, hence holds in the quotient $H^*(\mathbb{F}\ell_n)$. Apply the linear functional $\Lambda_{w/u}$ and use Theorem 4.3 together with Theorem 3.1 (which lets us write $N_{w/u}(\mathbf{p}_{\beta})$ for $\Lambda_{w/u}(Y^{\beta})$, independent of the chosen word of content β). $\square$

Explicitly, χ is computed from the one-variable-at-a-time recursion $h_{\alpha}(x_1, \dots, x_k) = \sum_{r=0}^a (Y_k - Y_{k-1})^r h_{\alpha-r}(x_1, \dots, x_{k-1})$, with $h_{\alpha}(\cdot) = [a=0]$.

Example 5.2. Take $n = 4$, $m = 2$, $\alpha = (0, 2, 0)$, so $h_{\alpha} = h_2(x_1, x_2)$. The recursion gives

$$h_2(x_1, x_2) = x_1^2 + x_1 x_2 + x_2^2 = Y_1^2 + Y_1(Y_2 - Y_1) + (Y_2 - Y_1)^2 = Y_2^2 - Y_1 Y_2 + Y_1^2.$$

So Theorem 5.1 reads $I_{(0,2,0)}(u, w) = N_{w/u}(22) - N_{w/u}(12) + N_{w/u}(11)$. Take $u = 1324$ and $w = 2413$. The Monk counts are $N_{w/u}(22) = 2$, $N_{w/u}(12) = 1$, $N_{w/u}(11) = 0$, so the formula gives $2 - 1 + 0 = 1$; and indeed $I_{(0,2,0)}(u, w) = 1$. Both naive comparisons fail on this instance: the sorted-word count alone gives $N_{w/u}(\mathbf{p}_{\alpha}) = N_{w/u}(22) = 2 \neq 1 = I_{\alpha}$, and the content-sum guess of Remark 2.3 gives $\binom{2}{(0,2,0)} N_{w/u}(22) = 2 \neq 1$ as well. The correction term is doing real work, and it is not a bijective correction: the one chain with word 12 is a different object from the two chains with word 22. What the identity says is that the *excess* $N_{w/u}(\mathbf{p}_{\alpha}) - I_{\alpha}(u, w)$ of non-increasing chains equals $N_{w/u}(12) - N_{w/u}(11)$. Finding a sign-reversing involution behind this is gap (2) of §8.

5.2 Samuel from Lenart–Sottile: the converse

Theorem 5.3. $\{h_\alpha : |\alpha| = m, \alpha \leq \delta\}$ is a $\mathbb{Z}$ -basis of $H^{2m}(\mathbb{F}\ell_n)$.

Proof, modulo one classical citation. Write $\psi_\alpha(f)$ for the coefficient of σ_{w_0} in $f \cdot h_\alpha$. By Lemma 4.2 with $u = e$ we have $h_\alpha = \sum_v I_\alpha(e, v) \sigma_v$, and by Poincaré duality $\langle \sigma_a \sigma_b, \sigma_{w_0} \rangle = \delta_{b, w_0 a}$. Taking $a = w_0 v$ gives

$$\psi_\alpha(\sigma_{w_0 v}) = I_\alpha(e, v).$$

Lemma 5 of math/0202090 states that $\psi_\alpha(f)$ is the coefficient of $x^{\delta-\alpha}$ in the normal form representative of f . Hence the matrix $M_{\alpha v} = I_\alpha(e, v)$, with α ranging over compositions $\leq \delta$ of m and v over $\{\ell(v) = m\}$, is the matrix of coefficients of $x^{\delta-\alpha}$ in $\mathfrak{S}_{w_0 v}$. Both index sets have the same cardinality, by the Lehmer-code bijection $v \mapsto \text{code}(v)$. The matrix is unitriangular with respect to the order induced by code, because $\mathfrak{S}_v = x^{\text{code}(v)} + (\text{lexicographically later monomials})$ — the classical triangularity of Schubert polynomials in the monomial basis (Lascoux–Schützenberger). Hence $\det M = \pm 1$. $\square$

Remark 5.4. The triangularity of Schubert polynomials is the one step of this paper that is cited rather than proved, and I have not re-derived it here. It is however *verified* as used: the matrices M were computed and inverted exactly over $\mathbb{Q}$ for every m at $n = 3, 4$, and every entry of every inverse is an integer (§7), which is unimodularity. The identity $\psi_\alpha(\sigma_{w_0 v}) = I_\alpha(e, v)$ derived above is also checked directly against instrument (A), and holds in 10/10, 106/106 and 1930/1930 instances at $n = 3, 4, 5$ — an independent confirmation of Lenart–Sottile’s Lemma 5.

Theorem 5.5. *There is a unique integer matrix $\eta = (\eta_{\beta\alpha})$, depending only on n and m , with $Y^\beta = \sum_\alpha \eta_{\beta\alpha} h_\alpha$ in $H^{2m}(\mathbb{F}\ell_n)$; and then for all $u \leq w$ with $\ell(w) - \ell(u) = m$,*

$$N_{w/u}(\mathbf{p}_\beta) = \sum_\alpha \eta_{\beta\alpha} I_\alpha(u, w).$$

Explicitly $\eta = \mathcal{N}\mathcal{I}^{-1}$, where $\mathcal{N}_{\beta v} = N_v(\mathbf{p}_\beta)$ and $\mathcal{I}_{\alpha v} = I_\alpha(e, v)$.

Proof. Existence, uniqueness and integrality of η are Theorem 5.3. Applying $\Lambda_{w/u}$ to $Y^\beta = \sum_\alpha \eta_{\beta\alpha} h_\alpha$ and using Theorem 4.3 gives the displayed identity. For the formula: $Y^\beta = \sum_v \mathcal{N}_{\beta v} \sigma_v$ by (2) at $u = e$, and $\sigma_v = \sum_\alpha (\mathcal{I}^{-1})_{v\alpha} h_\alpha$ by Lemma 4.2 at $u = e$. $\square$

Corollary 5.6 (answer to Q345). *The Lenart–Sottile chain formula and Samuel’s chain formula are related by a universal integer change of basis, in both directions, independent of u and w . Each determines the other, and each determines the structure constants $(c_{u,v}^w)_{\ell(v)=m}$. “Specialisation” is correct for the direction of Theorem 5.1 but understates the relationship; the accurate description is: the same functional $\Lambda_{w/u}$ expanded against a Monk-generated spanning set versus a Pieri-generated basis.*

6 Why no scalar factor can do it

One of the three outcomes anticipated before this session was “equal up to an explicit multinomial factor”. That is excluded structurally, not just numerically.

Proposition 6.1. *Restrict to a single block, $\alpha = a e_k$. Then $I_{a e_k}(u, w) \in \{0, 1\}$ for all u, w — the Pieri coefficient is multiplicity-free — whereas $N_{w/u}(k^a)$ is not bounded by 1: already in S_5 it takes the values 0, 1, 2, 3. Hence there is no function $c(m, \alpha, n)$ with $I_\alpha = c \cdot N(\mathbf{p}_\alpha)$.*

The multiplicity-freeness is Sottile's; here it is observed, not reproved: in all 6575 single-block instances at $n = 5$ the value of I_{ae_k} was 0 (5167 times) or 1 (1408 times), while $N_{w/u}(k^a)$ took the value 2 in 244 cases and 3 in 32. Since I and N take different values with ratio 1 in some instances and ratio $1/3$ in others, no single factor exists. The related guess $N(k^a) = a! I_{ae_k}$ fails in 870 of 4799 instances at $n = 5$.

7 Computational verification

Two independent instruments. **(A)** A Schubert instrument: Schubert polynomials by divided differences from $\mathfrak{S}_{w_0}$, structure constants by applying divided differences along a descent path. It contains no chains, no Monk rule and no Chevalley formula. **(B)** Two chain enumerators — the Samuel tensor $T_{w/u}$ in the root basis, and a Lenart–Sottile enumerator written from the increasing-chain definition alone. The LS enumerator never calls the Monk transfer matrices or $T_{w/u}$; the only code shared with (A) is the definition of S_n itself. Pure Python, exact integer arithmetic. Code at [proofs/code-1005-q345/](https://github.com/lenart/sottile).

Validating the Lenart–Sottile enumerator first.

check	$n = 3$	$n = 4$	$n = 5$
Prop. 3 (B–S): $\mathfrak{S}_u = \sum_{\gamma} x^{\delta}/x^{\gamma}$ vs (A)	6/6	24/24	120/120
Cor. 4: $I_{\alpha}(u, w) = \sum_v c_{u,v}^w I_{\alpha}(w_0 v, w_0)$	23/23	413/413	15839/15839

Of the Cor. 4 instances, 4, 206 and 12458 respectively have more than one term on the right, so the check is not a tautology on a singleton sum.

The load-bearing identities, against instrument (A). $\mathfrak{S}_u h_{\alpha} = \sum_w I_{\alpha}(u, w) \mathfrak{S}_w$ (Lemma 4.2): 25/25 at $n = 3$, 1009/1009 at $n = 4$. $\mathfrak{S}_u Y^{\beta} = \sum_w N_{w/u}(\mathbf{p}_{\beta}) \mathfrak{S}_w$ (eq. (2)): 25/25, 1009/1009. The two totals coincide because both loops run over the same index set $\{(m, \alpha \leq \delta, u, w)\}$; this is a shared denominator by construction, not an agreement between measurements.

The main theorem. Theorem 5.1: $I_{\alpha}(u, w) = \sum_{\beta} \chi_{\alpha}(\beta) N_{w/u}(\mathbf{p}_{\beta})$ — 25/25 at $n = 3$, 855/855 at $n = 4$, **53081/53081** at $n = 5$, zero disagreements.

Non-vacuity, measured. At $n = 3$ every χ_{α} has exactly one term, so $n = 3$ carries no information about this theorem at all: it is the degenerate boundary. At $n = 4$, 8 of the 23 functionals χ_{α} have 3 terms; at $n = 5$, 80 of 119 have more than one term, with term counts 1, 3, 5, 11 (multiplicities 39, 40, 20, 20). Likewise the comparison I_{α} versus $N(\mathbf{p}_{\alpha})$: of 603 instances at $n = 4$, the 218 with $\max \alpha \leq 1$ agree 218/218 as Theorem 2.2 predicts, and the 385 with $\max \alpha \geq 2$ disagree in 253 cases. Inequality $I_{\alpha} \leq N(\mathbf{p}_{\alpha})$: 0 violations at $n = 3, 4$.

Symmetry and transition. Theorem 3.1: all content classes constant, 20/20 at $n = 3$ and 603/603 at $n = 4$. Theorem 5.5: for each m the matrix $\mathcal{I}$ was inverted exactly over $\mathbb{Q}$; $\mathcal{I}^{-1}$ is integral in *every* case at $n = 3, 4$ and 5 — that is, $\det \mathcal{I} = \pm 1$ for all ten degrees $m = 1, \dots, 10$ at $n = 5$, which is the evidence for Theorem 5.3, the one step resting on a cited classical fact. η is integral throughout, and η computed *only* from $u = e$ reproduces $N_{w/u}(\mathbf{p}_{\beta})$ for all (u, w) : 16 + 12 + 4 instances at $n = 3$, 174 + 378 + 420 + 285 + 126 + 28 = 1411 at $n = 4$, and 16 442 at $n = 5$, zero failures.

The $n = 5$ run is a *sample*, and the sampling is stated rather than hidden: for each m the universality loop tested one in every seven eligible pairs (u, w) . At $m = 10$ there is a single eligible pair and the sample skipped it, so that degree contributes 0/0 to the universality count — its unimodularity claim is unaffected. The dimensions $\dim H^{2m}(\mathbb{F}\ell_5) = 1, 4, 9, 15, 20, 22, 20, 15, 9, 4, 1$ are Mahonian, as they must be, and η is far from trivial at $n = 5$: the number of multi-term rows rises 0, 3, 12, 27, 47, 70, 89, 99, 82, 0 out of 4, 10, 20, 35, 56, 84, 120, 165, 220, 286.

Refutations. The content-sum guess $I_\alpha \stackrel{?}{=} \sum_{\text{content}(\mathbf{p})=\alpha} N_{w/u}(\mathbf{p})$ fails 8/20 at $n = 3$ and 481/603 at $n = 4$. The block-factorial guess $N(k^a) \stackrel{?}{=} a! I_{ae_k}$ fails 2/4, 48/168, 870/4799 at $n = 3, 4, 5$.

7.1 Planted controls

The LS enumerator asserts three things — the label range $i \leq k < j$, the second coordinate $b = u(i)$, and strict lexicographic increase on pairs — so it was given one control per assertion. Three of six fired; the three silences are each explained, and two of them are findings.

- **Fires.** Widening the range to $i \leq k \leq j$; dropping the lexicographic condition entirely; ordering by first coordinate only; reversing to lexicographically decreasing — each produces a negative exponent in $x^{\delta-\alpha}$, i.e. breaks Prop. 3 immediately. Ordering by (b, k) instead of (k, b) gives 10 mismatches at $n = 4$. So the instrument is sensitive to precisely the feature under test.
- **Silent, and why.** (i) Using $b = u(j)$ instead of $b = u(i)$ changes nothing — which *confirms* the parenthetical remark of `math/0202090` that “all the constructions and statements still hold if we uniformly set $b = u(j) = w(i)$ ”. (ii) Weakening strict to non-strict lexicographic order changes nothing, by Remark 2.4. (iii) Likewise requiring only weak increase of b within blocks. In each case the control is vacuous rather than missed.

My first explanation of silence (ii) was wrong, and the controls caught it: I claimed a label can never repeat in a chain, and exhaustive enumeration found 598 chains at $n = 4$ that repeat one. The argument is valid only for *consecutive* repeats — which is exactly what the relaxation admits, so the conclusion survives — because a cover moves two values and can carry b back to the left of position k at a later step.

8 What is proved, and what is not

Proved here, self-contained. Theorems 2.1, 2.2, 3.1, Corollary 3.2, Remark 2.4, and Theorem 5.1. Theorem 5.1 — the direction Q345 asked about — rests only on (2) (proved 2026-10-04 from Monk’s rule), Lemma 4.2, and the substitution $x_p = Y_p - Y_{p-1}$.

Cited. (i) Sottile’s Pieri formula (3), quoted from `math/0202090` (**deep-read**); [2] itself is unread by me. It is used only in Lemma 4.2, which is verified independently at $n = 3, 4$. (ii) Lemma 5 of `math/0202090` and the triangularity of Schubert polynomials, used only in Theorem 5.3 and hence only for the converse direction, Theorem 5.5; unimodularity is verified exactly at $n = 3, 4$.

Gaps, precisely.

1. Verification is exhaustive for $n \leq 4$ throughout, and at $n = 5$ for the principal identity, Prop. 3, Cor. 4, the single-block statistics and the unimodularity of $\mathcal{I}$. The one place a bound was imposed is the $n = 5$ universality loop for η , which ran on a stated 1-in-7 sample of eligible pairs; nothing beyond $n = 5$ was attempted anywhere.
2. No closed form for $\chi_\alpha(\beta)$ is given beyond the recursion. The observed term counts 1, 3, 5, 11 at $n = 5$ are suggestive but unexplained; a combinatorial (ideally sign-reversing-involution) description of χ would upgrade Theorem 5.1 from a substitution to a bijection, and is the natural next question.
3. Proposition 6.1 *observes* Pieri multiplicity-freeness rather than proving it.
4. Everything is for the finite flag variety. Samuel's V is defined for S_∞ and his theorem holds there; h_α and δ are n -dependent, so Theorem 5.5 is genuinely a statement about S_n . Theorem 5.1 is stable in n , since the identity $h_\alpha = \sum_\beta \chi_\alpha(\beta) Y^\beta$ is an identity of polynomials.

References

- [1] C. Lenart and F. Sottile, *Skew Schubert polynomials*, `arXiv:math/0202090`. Read at source (`LATEX` source, compiled locally); Thm 2 = `thm:skew` p.4, Prop. 3 = `prop:Schub-constr`, Cor. 4 = `cor:identity`, Lemma 5 = `lem:linear`, Pieri = eq. E:Pieri.
- [2] F. Sottile, *Pieri's formula for flag manifolds and Schubert polynomials*, Ann. Inst. Fourier 46 (1996). *Not read by me*; cited only as quoted in [1].
- [3] M. Samuel, answer to MathOverflow 313951. Proved in `proofs/2026-10-04-c4-samuel-chain-formula.tex`.
- [4] N. Bergeron and F. Sottile, as cited in [1] for Prop. 3.